quick, and low-power control of smart devices, without the need of a router. It helps in communication without any internet connectivity.

In its one-way communication mode, the protocol lets one ESP32 board to send data to another ESP32 board (see Fig. 6). This configuration is easy to implement and great for sending data like sensor readings or on/off commands to control GPIOs from one board to the other. One ESP32 board can also send the same or different commands to different ESP32 boards. This configuration is

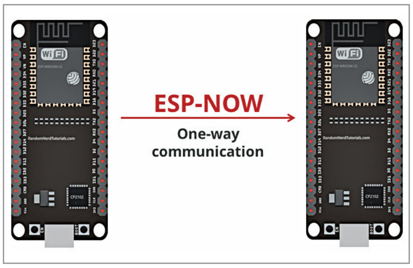

Fig. 6: ESP-NOW one-way communication protocol

```
sketch_apr12a §
#include <esp_now.h>
#include <WiFi.h>
#include <Adafruit_MPU6050.h>
#include <Adafruit_Sensor.h>
#include <Wire.h>

Adafruit_MPU6050 mpu;

int delay_sec = 5;
int count;

// REPLACE WITH YOUR RECEIVER MAC Address
uint8_t broadcastAddress[] = {0x9C, 0x9C, 0x1F, 0xC8, 0x57, 0x84};

// Structure example to send data
// Must match the receiver structure
typedef struct struct_message {
  char a[32];
  int b;
  float c;
  bool d;
} struct_message;

// Create a struct_message called myData
struct_message myData;

esp_now_peer_info_t peerInfo;

// callback when data is sent
void OnDataSent(const uint8_t *mac_addr, esp_now_send_status_t status) {
  Serial.print("\r\nLast Packet Send Status:\t");
  Serial.println(status == ESP_NOW_SEND_SUCCESS ? "Delivery Success" : "Delivery Fail");
}
```

Fig. 8: Transmitter code snippet setting the ESP NOW configuration

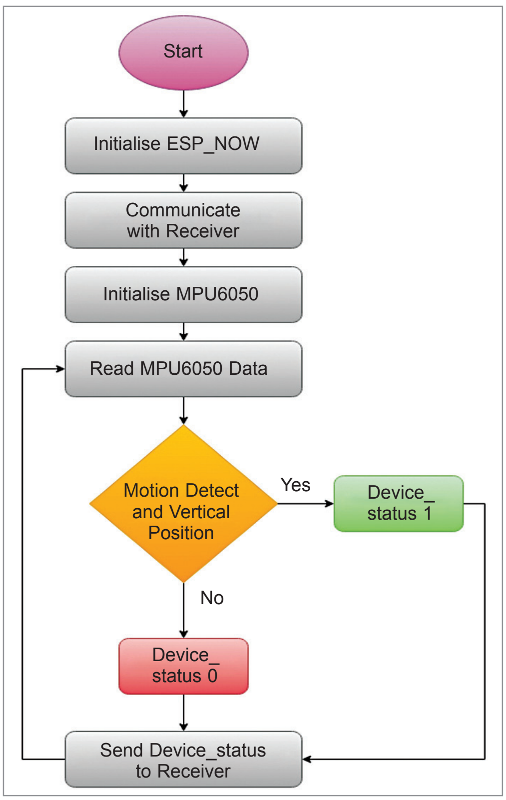

Fig. 7: Flow chart of transmitter code

ideal to build something like a remote control.

To communicate via ESP-NOW, you need to know the Mac address of the ESP32 receiver. Each ESP32 has a unique Mac address, which helps us identify each board for sending data to it using ESP-NOW.

Software

Separate software codes are needed for the transmitter and the receiver, which are created using Arduino IDE. So, first install the IDE and then install the ESP board. Then open and install the Addafruit mpu6050 and ESP-NOW libraries. Select the board as ESP32 and prepare the code for the transmitter. Here the MPU 6050 6-axis gyro sensor and accelerometer are used to sense the motion of your hand ironing the clothes and get to know whether you are using the iron or not.

The working code's flow chart is shown in Fig. 7. First, the device establishes the connection to the receiver using ESP-NOW and then it senses the motion and gives the signal to the receiver to turn the iron on or off. So, in the code first you need to initialise the ESP-NOW library and then the Addafruit mpu6050 library. Next, you need to configure the Mac address to the receiver to pair and communicate with.

Fig. 8 shows the snippet of the source code of transmitter. The loop function in code checks the motion and sends a signal to the receiver. After completing the code, upload it after selecting the port name and board in IDE.

To prepare the code for receiver, first initialise the ESP-NOW library and then check the incoming message. In the loop, turn the relay